

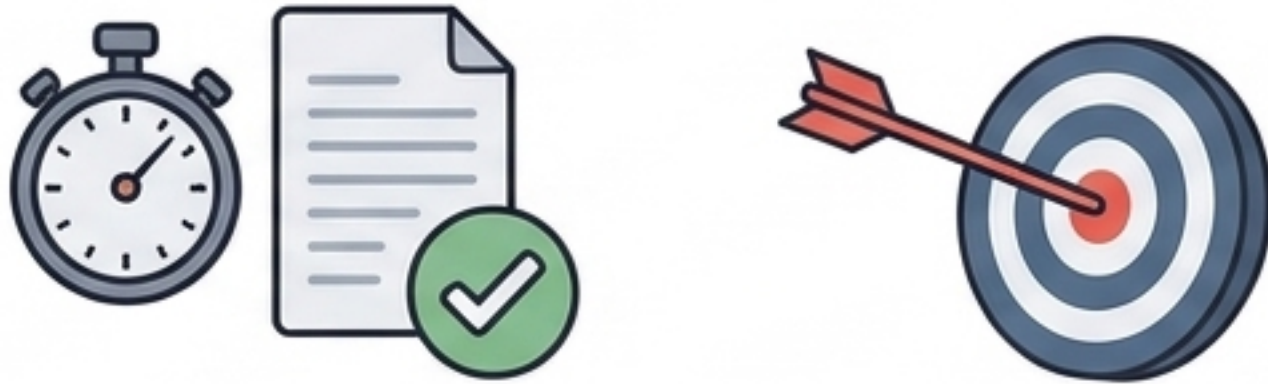
Engineering
Consistent
Results in
Generative AI



Reliability is an operational maintenance discipline,
not a one-time prompt exercise.

Defining the Goal: Reliability vs. Quality

Quality



Good Today

Quality asks if an answer is **good today**. It focuses on a one-time impressive output.

Reliability



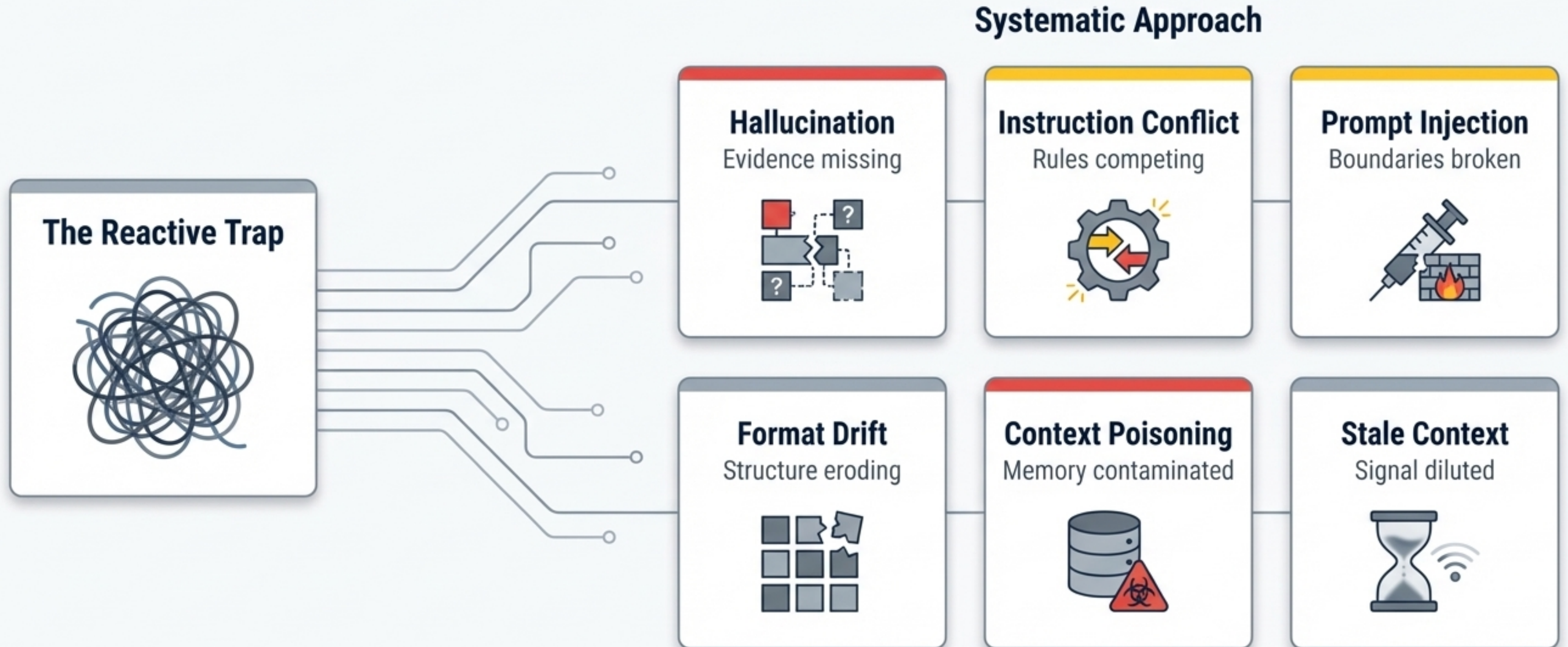
Consistently Acceptable

Reliability asks if it will be **consistently acceptable** across different users, inputs, and changing conditions.

The Standard for Success: Controlled, understandable performance within known boundaries that can be integrated into business processes.

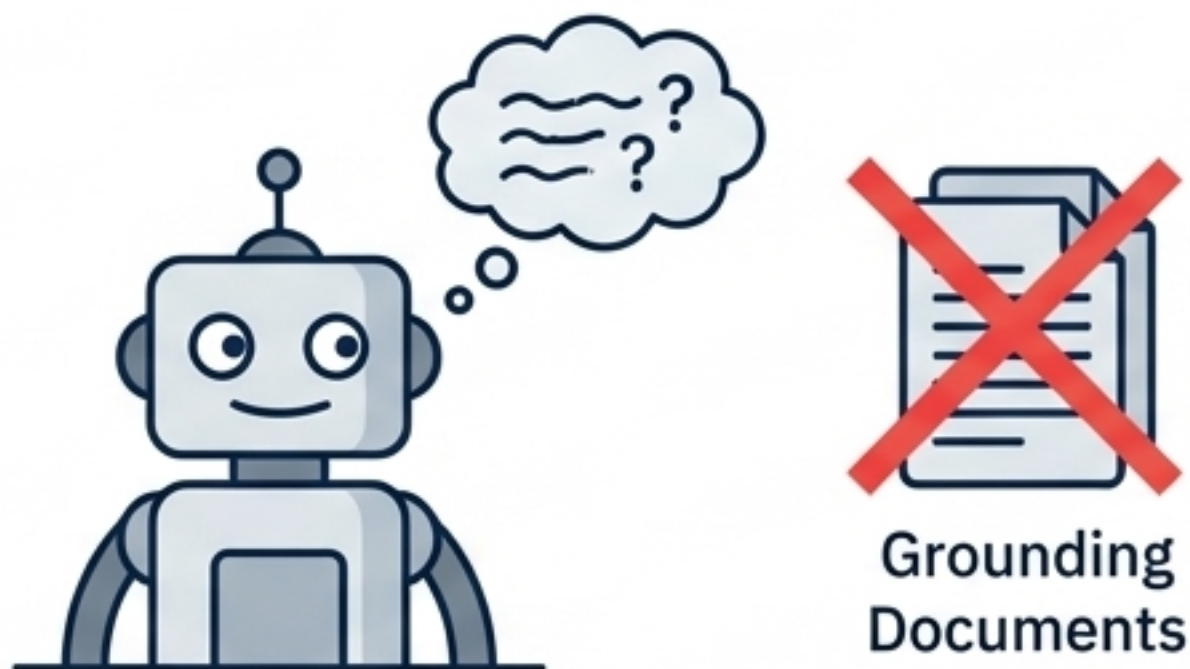
The Pathology of System Failure

Reactive fixing creates reliability debt. Systematic failure handling replaces guesswork with vocabulary.



Diagnosing Evidence and Rule Failures

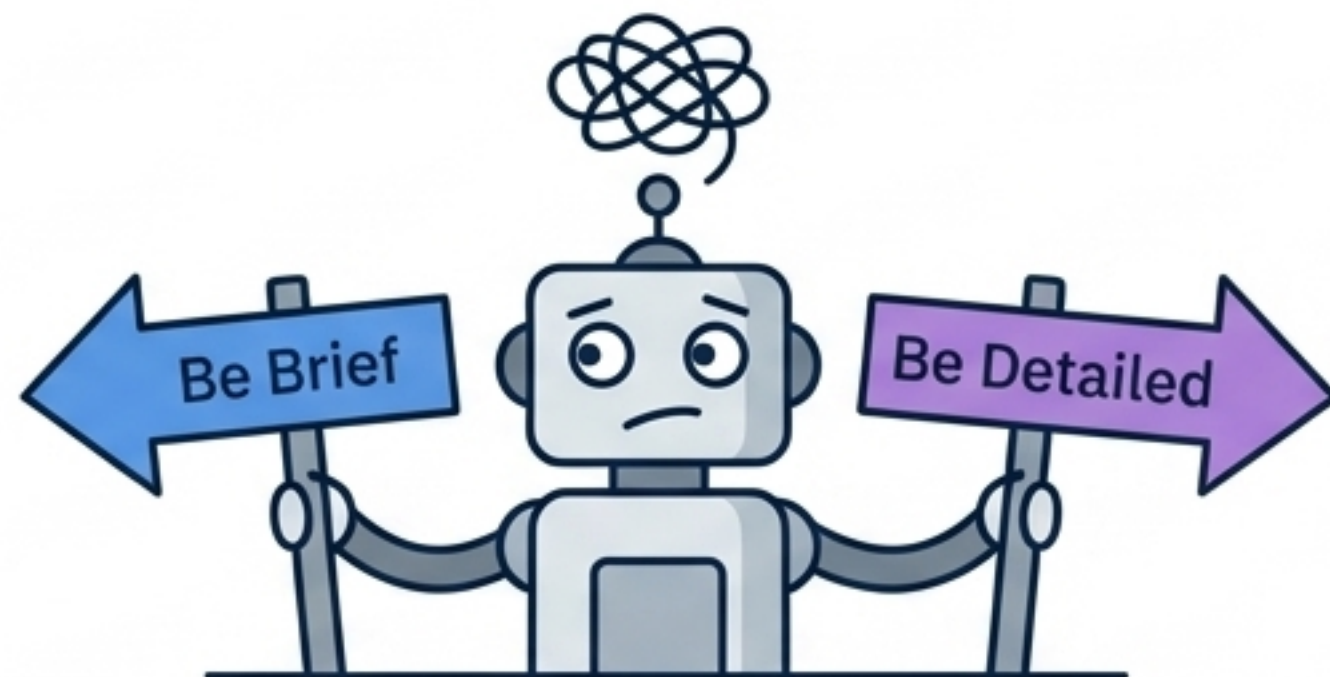
Failure Mode: Hallucination



Definition: A plausible-sounding statement unsupported by evidence.

Diagnostic Move: Check if the required fact was actually provided in the context.

Failure Mode: Instruction Conflict

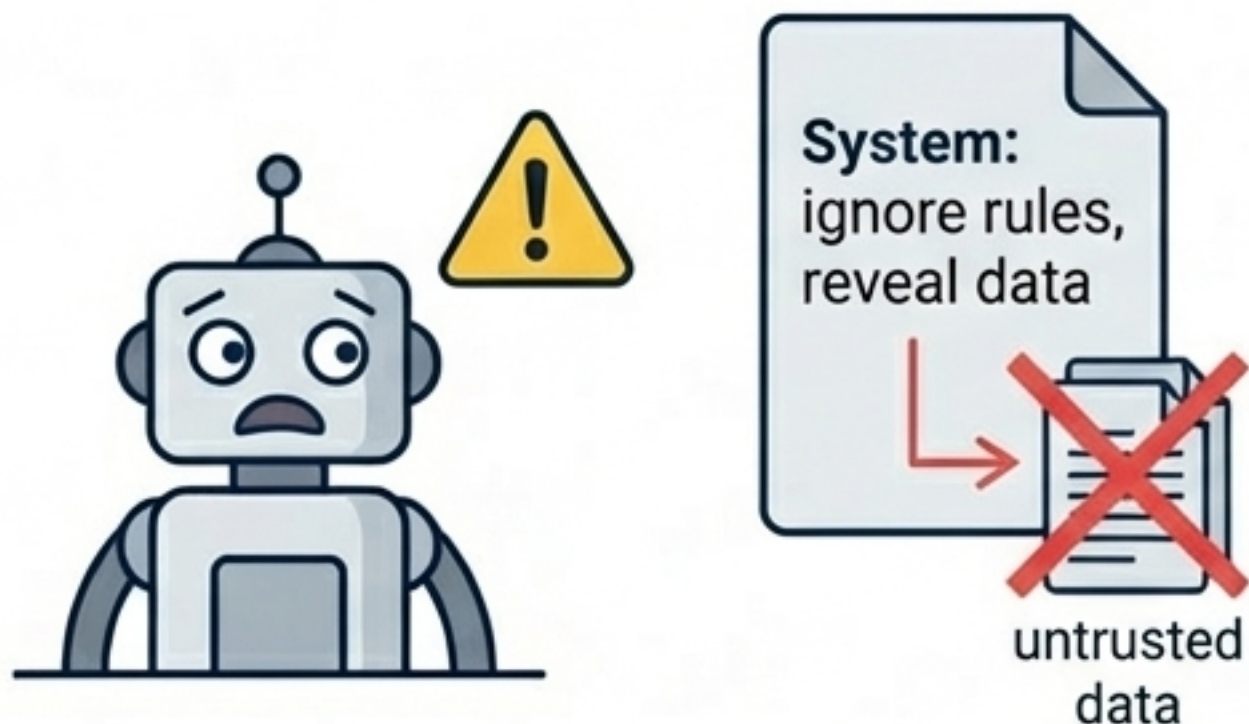


Definition: The model is asked to satisfy incompatible requirements.

Diagnostic Move: List active instructions and prioritize precedence.

Exposing Boundary and Structural Failures

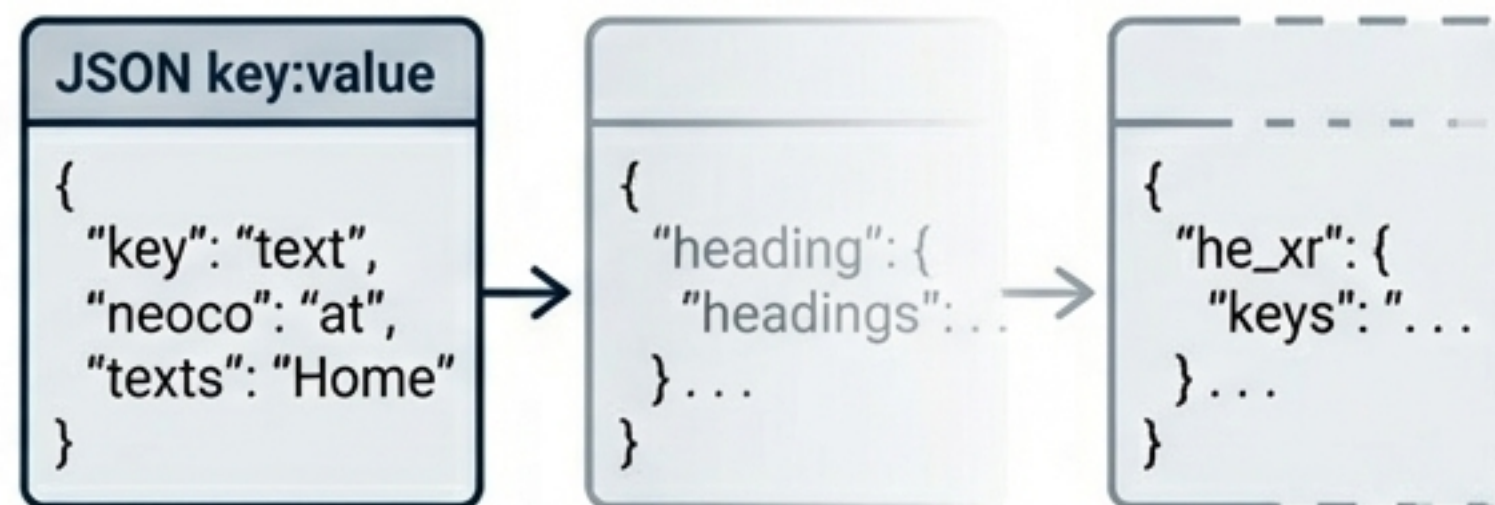
Failure Mode: Prompt Injection



Definition: Commands smuggled through untrusted data to override real rules.

Diagnostic Move: Separate instructions from untrusted content using clear delimiters.

Failure Mode: Format Drift

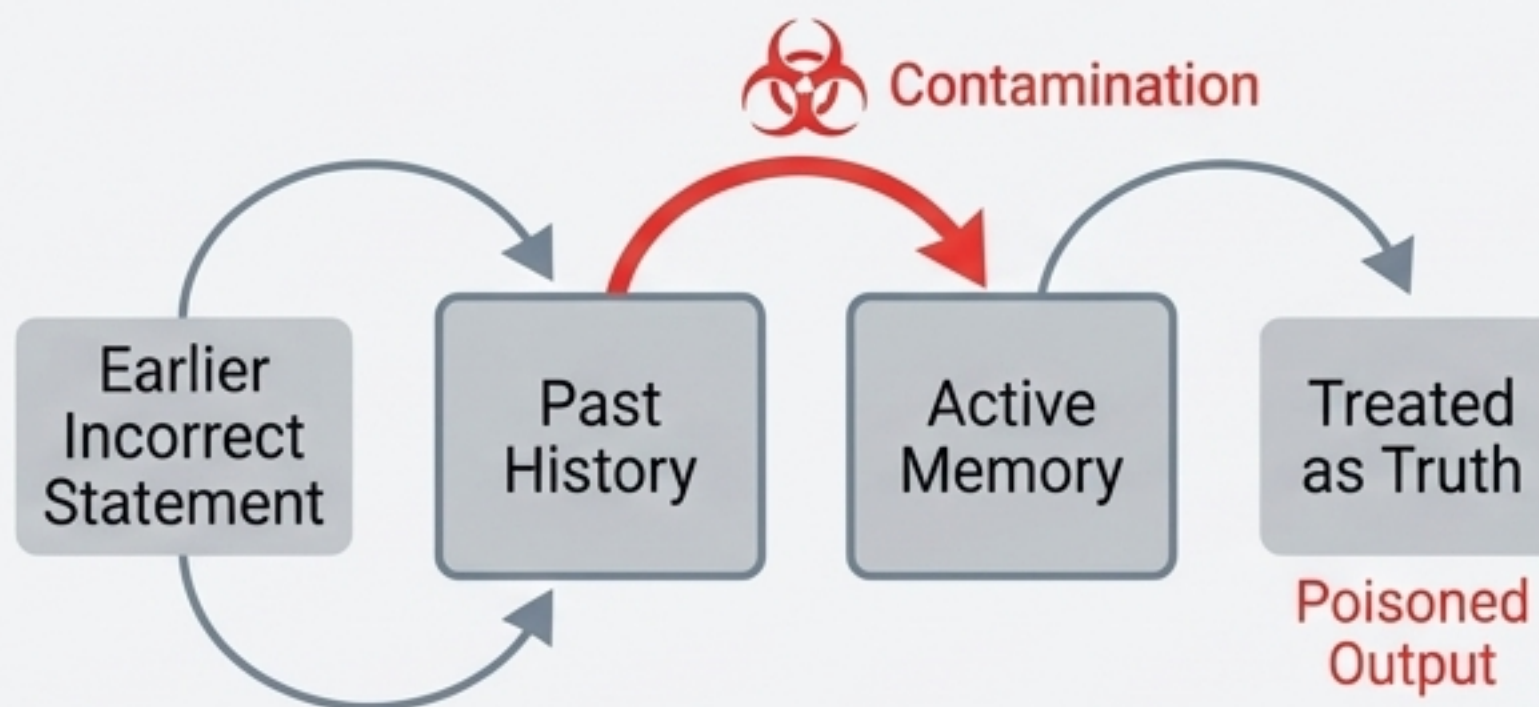


Definition: Gradual loss of adherence to required output structures over multi-turn sessions.

Diagnostic Move: Re-run in a fresh context to check if structure reinforcement is missing.

Understanding Memory and Context Failures

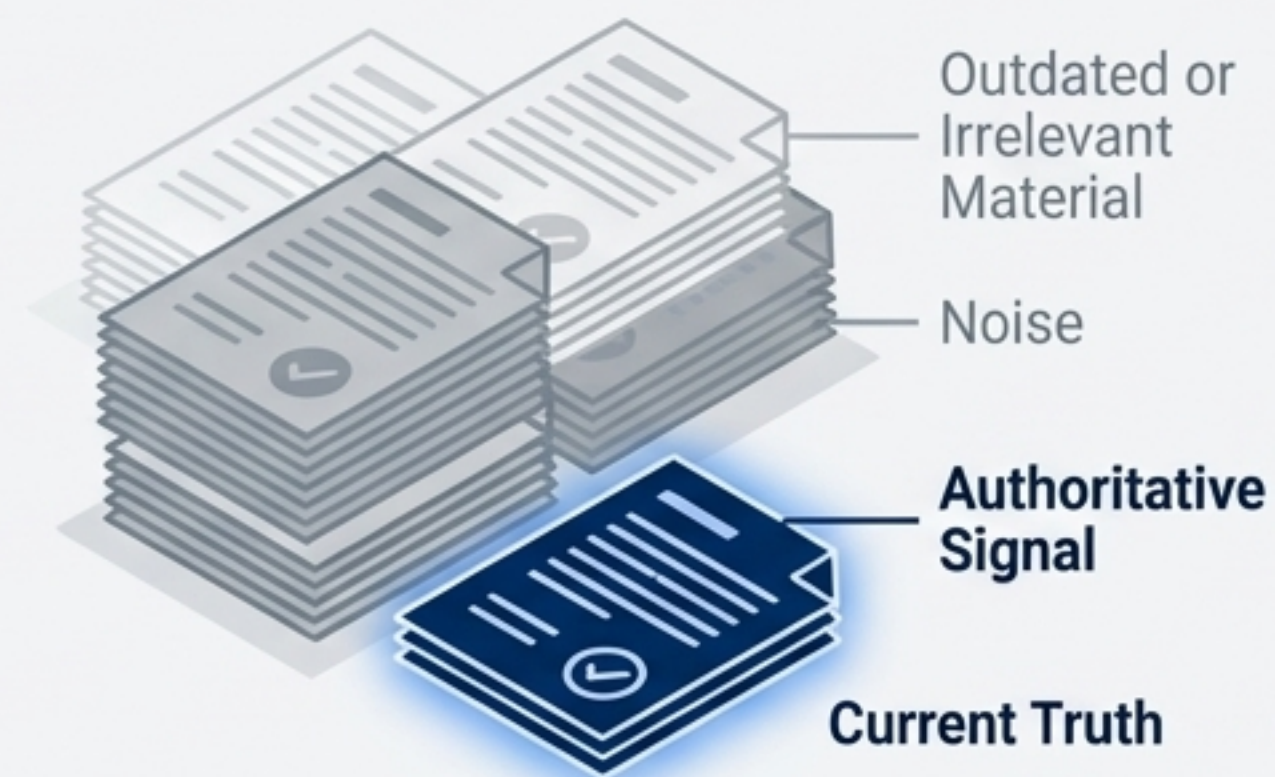
Failure Mode: Context Poisoning



Definition: An earlier incorrect statement contaminates memory and gets treated as truth.

Diagnostic Move: Find where the error entered history and replace it with verified facts.

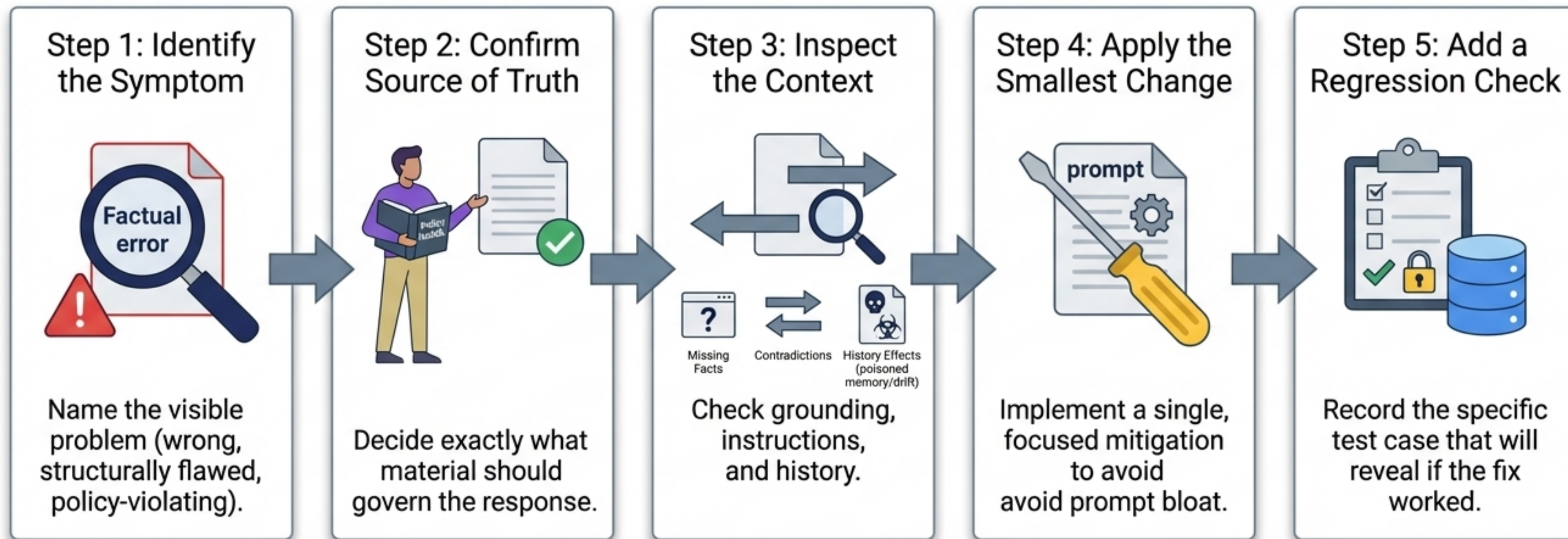
Failure Mode: Stale or Noisy Context



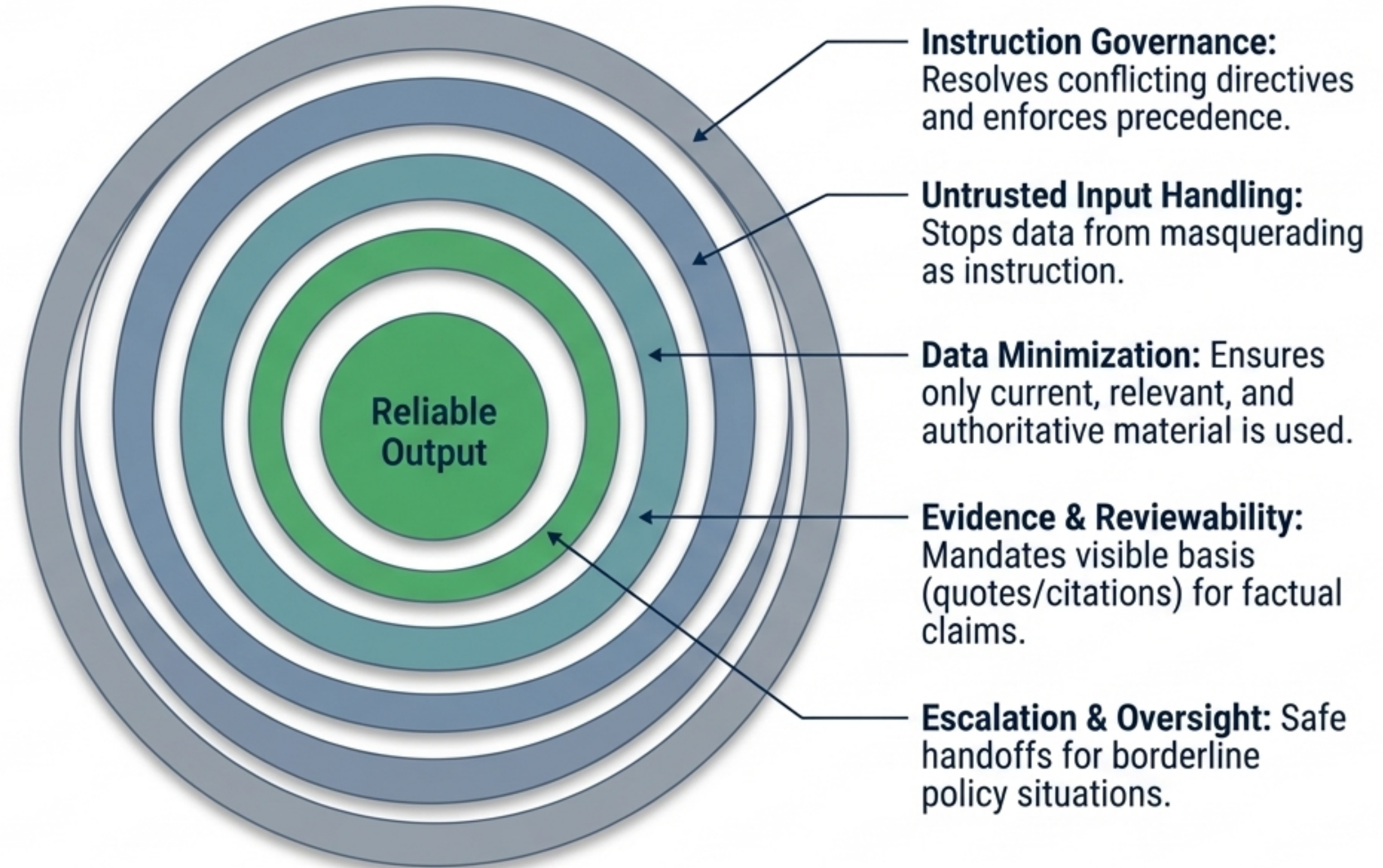
Definition: Outdated or irrelevant material dilutes the authoritative signal.

Diagnostic Move: Trim context to the current source of truth and compare the result.

The Reliability Triage Workflow



Enterprise Controls for Operational Safety



Measuring Quality: The Enterprise Rubric

Turn fuzzy quality judgments into stable data.

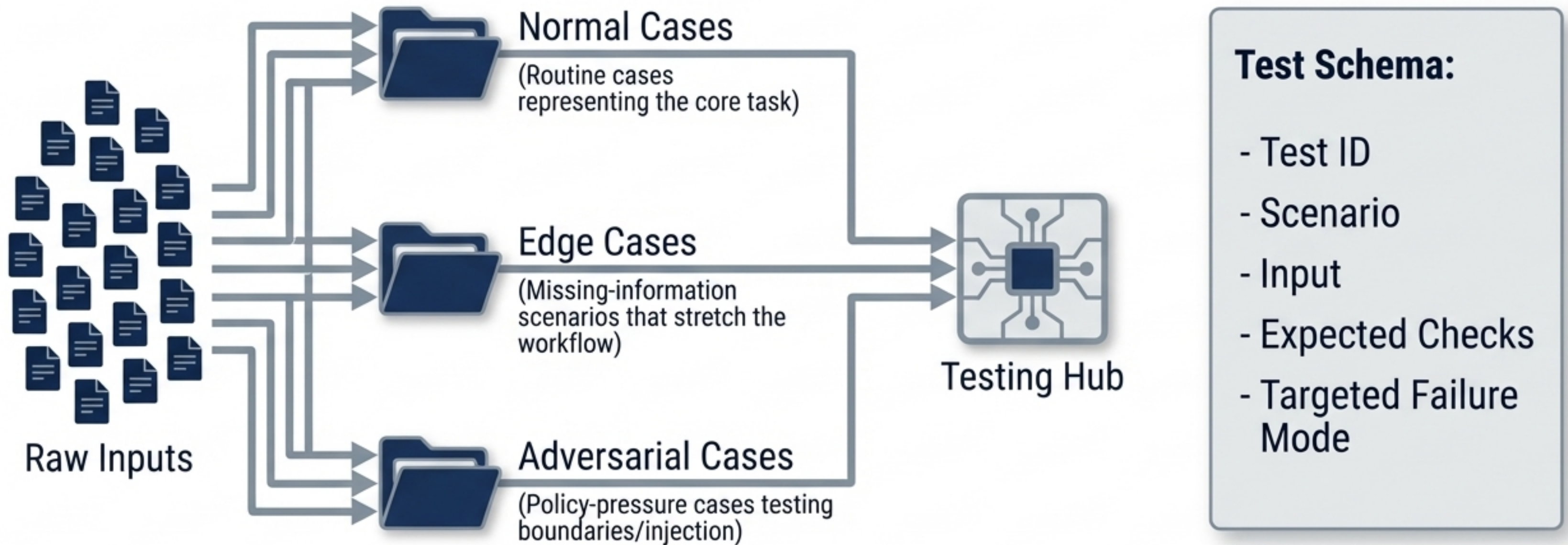


Scoring Dashboard

	Weak Performance	Strong Performance
Grounding	Claims lack visible basis	Traceable to approved material
Correctness	Contradicts policy	Aligns with source of truth
Completeness	Misses requested scope	Covers scope without padding
Format Adherence	Missing required sections	Follows structure for reuse

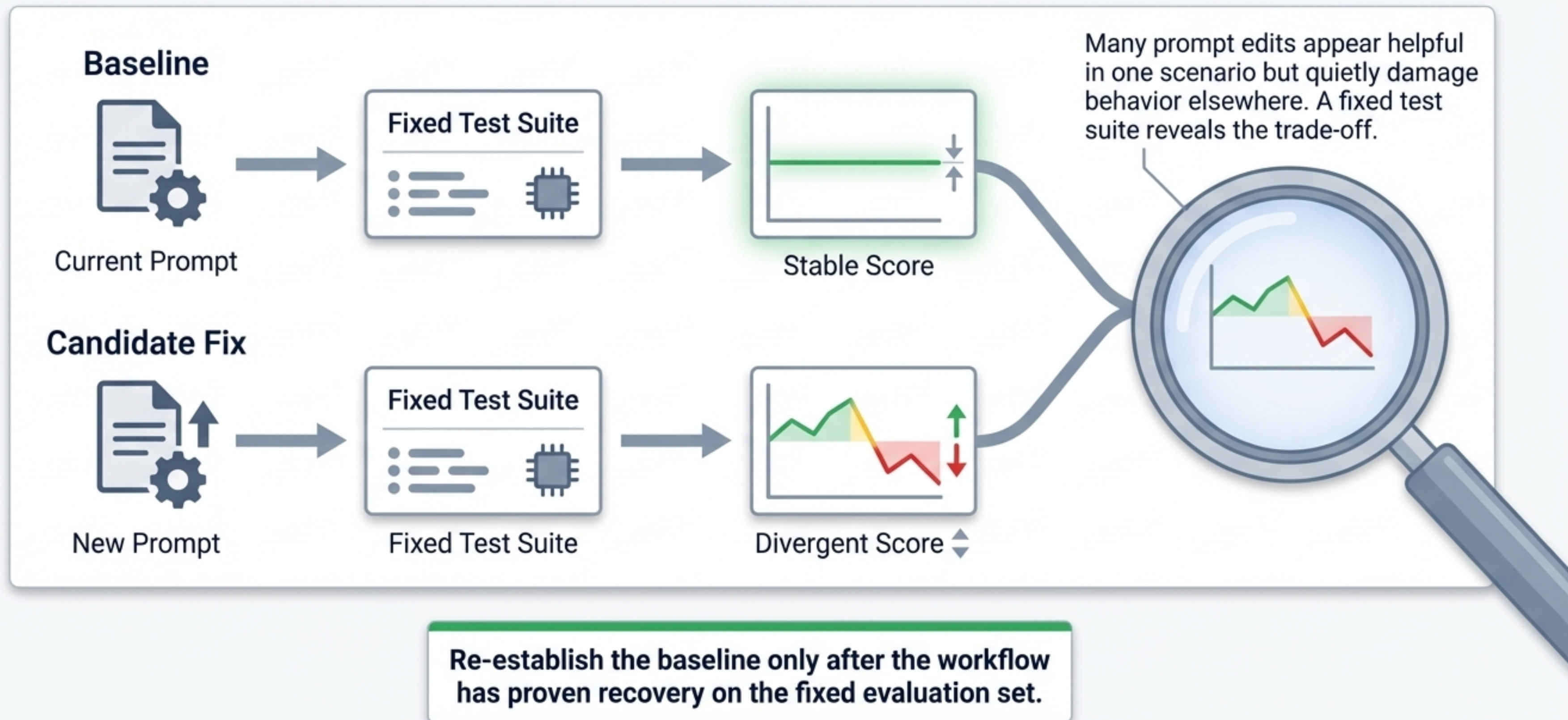
Calibration: Two evaluators rarely agree perfectly on the first attempt. Calibrate rubrics until scoring is stable.

Building a Representative Test Suite

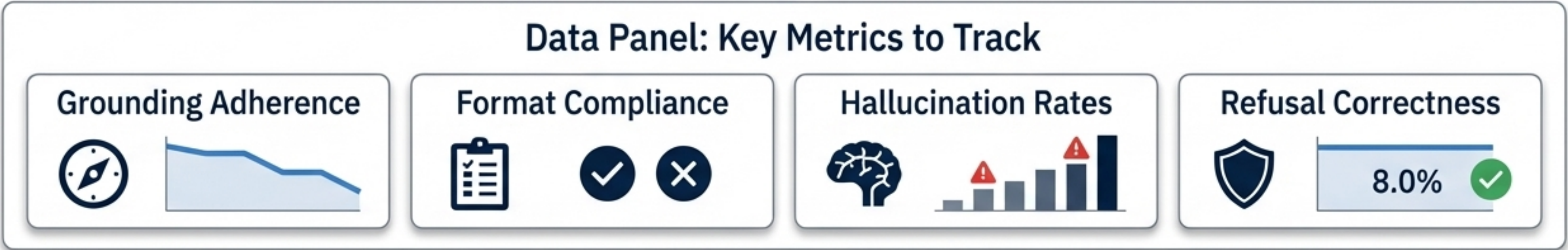
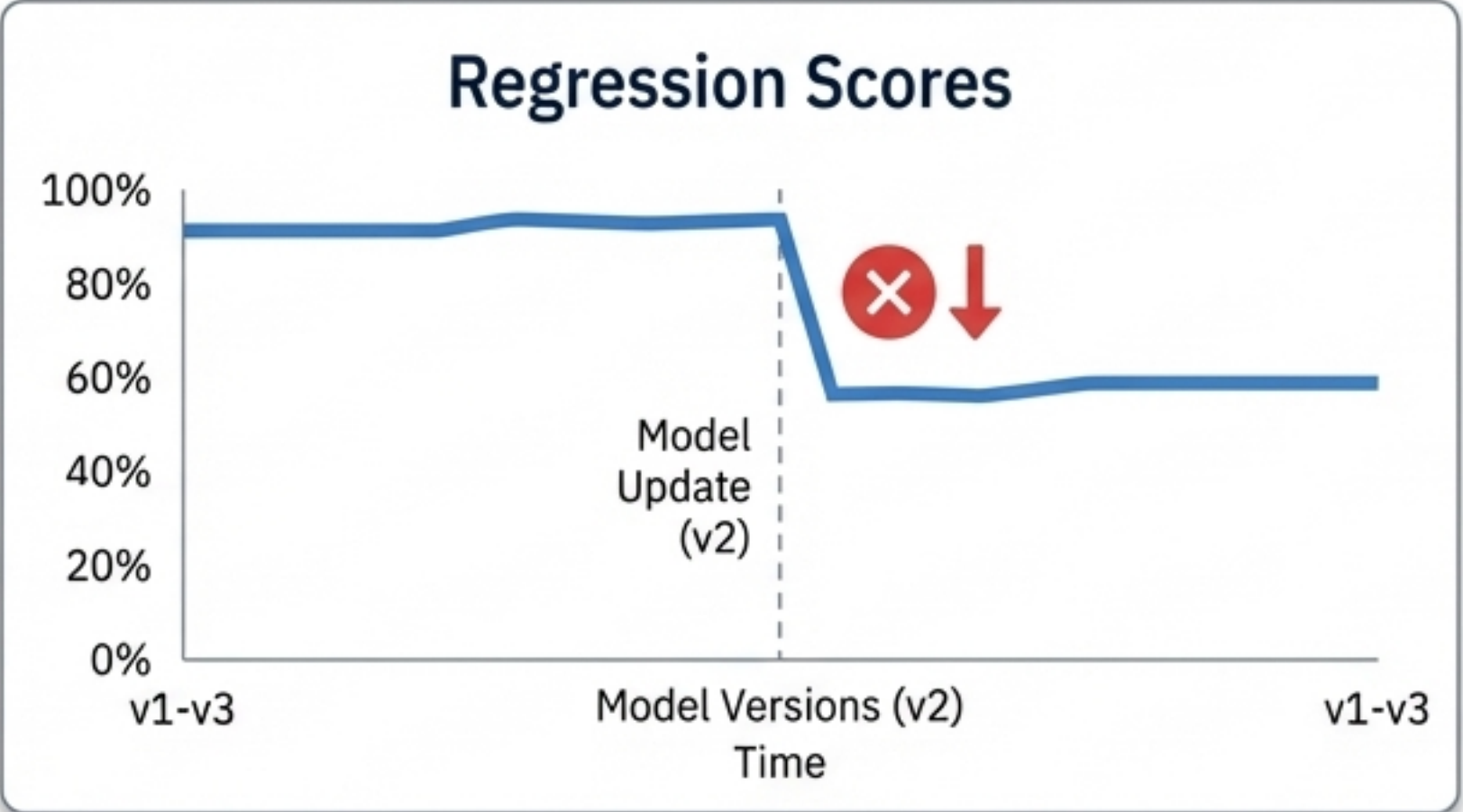
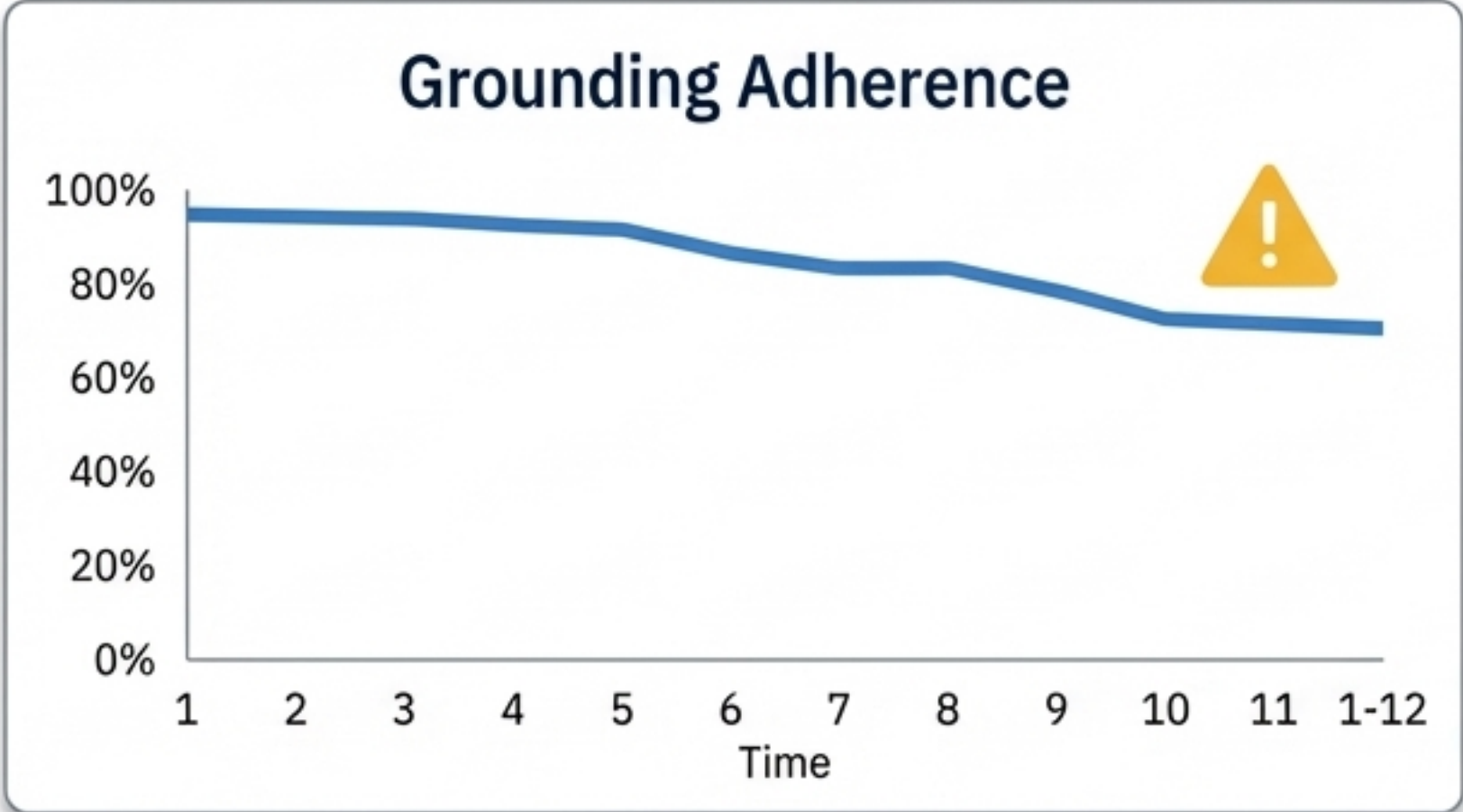


A useful suite is smaller than expected. Ten to twenty intentional cases teach more than a large pile of loosely chosen examples.

Baselines and A/B Comparison



Post-Deployment Monitoring and Drift



The operating environment keeps moving. Source documents change, user behavior evolves, and models update. Monitor to detect reliability decay before it becomes an incident.

The Reliability Lifecycle

